Check for updates

ADVANCING EARTH AND SPACE SCIENCE

JGR Solid Earth

RESEARCH ARTICLE
10.1029/2021JB023249

Earthquake Phase Association Using a Bayesian Gaussian Mixture Model

Weiqiang Zhu[1], Ian W. McBrearty[1], S. Mostafa Mousavi[1], William L. Ellsworth[1], and Gregory C. Beroza[1]

[1]Department of Geophysics, Stanford University, Stanford, CA, USA

Special Section:
Machine learning for Solid Earth observation, modeling and understanding

Key Points:

- We proposed an new approach to solve phase association as an unsupervised clustering problem using the Bayesian Gaussian Mixture Model
- We used the multivariate Gaussian distribution to represent both phase arrival time and amplitude to improve association
- Our unsupervised method is fast without the need for conventional grid-search or supervised training

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

W. Zhu,
zhuwq@stanford.edu

Citation:

Zhu, W., McBrearty, I. W., Mousavi, S. M., Ellsworth, W. L., & Beroza, G. C. (2022). Earthquake phase association using a Bayesian Gaussian Mixture Model. *Journal of Geophysical Research: Solid Earth*, *127*, e2021JB023249. https://doi.org/10.1029/2021JB023249

Received 18 SEP 2021
Accepted 27 MAR 2022

Author Contributions:

Conceptualization: Weiqiang Zhu, Ian W. McBrearty, S. Mostafa Mousavi, William L. Ellsworth, Gregory C. Beroza
Data curation: Weiqiang Zhu
Formal analysis: Weiqiang Zhu
Funding acquisition: Gregory C. Beroza
Investigation: Weiqiang Zhu
Methodology: Weiqiang Zhu, Ian W. McBrearty, S. Mostafa Mousavi, William L. Ellsworth, Gregory C. Beroza
Project Administration: Gregory C. Beroza
Resources: Weiqiang Zhu
Software: Weiqiang Zhu

© 2022. American Geophysical Union. All Rights Reserved.

Abstract Earthquake phase association algorithms aggregate picked seismic phases from a network of seismometers into individual sesimic events and play an important role in earthquake monitoring and research. Dense seismic networks and improved phase picking methods produce massive seismic phase datasets, particularly for earthquake swarms and aftershocks occurring closely in time and space, making phase association a challenging problem. We present a new association method, the Gaussian Mixture Model Association (GaMMA), that combines the Gaussian mixture model with earthquake location, origin time, and magnitude estimation. We treat earthquake phase association as an unsupervised clustering problem in a probabilistic framework, where each earthquake corresponds to a cluster of P and S phases with a hyperbolic moveout of arrival times and a decay of amplitude with distance. We use the multivariate Gaussian distribution to model the collection of phase picks of an event; and the mean of the multivariate Gaussian distribution is given by the predicted arrival time and amplitude from the causative event. We carry out the pick assignment to each earthquake and determine earthquake source parameters (i.e., earthquake location, origin time, and magnitude) under the maximum likelihood criterion using the Expectation-Maximization algorithm. The GaMMA method does not require typical association steps of other algorithms, such as grid-search or supervised training. The results for both synthetic tests and for the 2019 Ridgecrest earthquake sequence show that GaMMA effectively associates phases from a temporally and spatially dense earthquake sequence while producing useful estimates of earthquake location and magnitude.

Plain Language Summary Earthquakes are monitored by seismic networks consisting of several to hundreds of seismometers. An earthquake detection workflow usually has two important steps: detecting/picking seismic phases at each seismometer and associating picked phases across multiple seismometers in a network. Deep-learning-based phase pickers have greatly improved phase detection performance and can automatically generate many more seismic phases than conventional algorithms. These massive numbers of automatic phase picks pose a challenge for the phase association task. We have developed a new phase association method using a Bayesian Gaussian Mixture Model. We treat the phase association problem as a unsupervised clustering problem meaning that we aim to cluster detected phases into different groups based on individual earthquakes that produce these phases. The Gaussian mixture model makes it easy to consider multiple phase parameters, such as phase arrival time, phase amplitude, phase picking quality score, and phase type, to improve phase association. We test our method on both synthetic data and the 2019 Ridgecrest earthquake. The results show that our method can effectively associate phases from a temporally and spatially dense earthquake sequence and generate a more complete earthquake catalog than catalogs created using conventional methods.

1. Introduction

Earthquake catalogs are fundamental products that are widely used in seismology to study and model various aspects of seismicity. Extensive efforts have been made to generate more complete catalogs with many more smaller earthquakes and more precise location and magnitude estimates. These high-resolution high-precision catalogs have the potential to reveal relationships among earthquakes and illuminate active structures that would otherwise remain hidden (Beroza et al., 2021; Hauksson et al., 2012; Park et al., 2020; Ross, Trugman, et al., 2019; Tan et al., 2021; Waldhauser & Schaff, 2008; Yoon et al., 2015). A standard earthquake-monitoring workflow from seismic waveforms to earthquake catalogs includes several tasks, including: phase detection and picking (Allen, 1978), phase association (Yeck et al., 2019), earthquake location (Klein, 2002), and magnitude estimation (Richter, 1935).

Supervision: William L. Ellsworth, Gregory C. Beroza
Validation: Weiqiang Zhu, Gregory C. Beroza
Visualization: Weiqiang Zhu
Writing – original draft: Weiqiang Zhu
Writing – review & editing: Weiqiang Zhu, Ian W. McBrearty, S. Mostafa Mousavi, William L. Ellsworth, Gregory C. Beroza

The phase picking step detects seismic phases such as P-wave and S-wave phases at each seismic station. The phase association step aggregates these phases from multiple stations of a seismic network into separate groups associated with each earthquake. Earthquake location and magnitude are then estimated from the associated phase information such as arrival time and amplitude. The resulting catalog can be further enhanced through template matching (Gibbons & Ringdal, 2006; Peng & Zhao, 2009; Shelly et al., 2007) or subspace projection (Barrett & Beroza, 2014; Harris & Dodge, 2011) that use the detected earthquakes as templates to re-scan the waveforms and detect small earthquakes with similar waveforms.

The phase picking step has been significantly improved by deep-learning-based pickers that learn from manual picks labeled by analysts to detect millions of phase picks from raw seismic waveforms (Mousavi et al., 2020; Ross et al., 2018; W. Zhu & Beroza, 2018). The rapidly growing volume of automatic picks and the ongoing growth of seismic networks makes developing effective phase association methods crucial; yet, phase association has not received the attention that has been devoted to other earthquake monitoring tasks. Association methods based on back-projection are most commonly used in classic earthquake monitoring systems such as Hydra (Patton et al., 2016), Earthworm (Friberg et al., 2010), and SeisComP3 (Weber et al., 2007). These approaches usually deploy a grid-search, and back-project phase picks based on the expected moveout with distance. An earthquake and its initial location is declared based on the number of phase picks inside a spatial grid. Although back-projection-based association is robust and effective, its performance is limited for dense earthquake sequences when earthquakes occur so closely in time and space that interpreting the maxima resulting from back-projection becomes problematic. Studies have continued to focus on improving the grid-search and back-projection approach for different scenarios (Arora et al., 2013; Draelos et al., 2015; Gibbons et al., 2016; Zhang et al., 2019). Specifically, both Net-VISA (Arora et al., 2013) and PEDAL (Draelos et al., 2015) incorporated a probabilistic framework to add prior information from historical catalogs and consider the uncertainty of phase picks, velocity models, and grid sizes. These methods rely on a complex combination of probabilistic models and many hyper-parameters that control performance and accuracy. Several new approaches have been proposed to solve the earthquake phase association using: graph theory (McBrearty, Gomberg, et al., 2019), the RANSAC algorithm (L. Zhu et al., 2021; Woollam et al., 2020), and deep learning (Dickey et al., 2020; McBrearty, Delorey, & Johnson, 2019; Ross, Yue, et al., 2019). When combined with the rapid development of phase picking methods, better association methods have the potential to improve significantly the overall performance of earthquake monitoring pipelines.

We propose an association method based on a Bayesian Gaussian Mixture Model (GMM), which is an unsupervised machine learning method for clustering (Bishop, 2006) that has been widely used in different research fields, such as image processing (Permuter et al., 2006), speech recognition (Reynolds & Rose, 1995), and earthquake studies (Ross et al., 2020; Seydoux et al., 2020). Earthquake phase association can be treated as an unsupervised clustering problem, with groups of phase picks, in time and space, arising from a discrete set of earthquake origins. We combine the GMM with earthquake location, origin time, and magnitude estimation, so that the new Gaussian Mixture Model Association (GaMMA) method can cluster phase picks based on the physical constraints of arrival time moveout and amplitude decay with distance. Phase amplitude information is often neglected because it can be difficult to account in conventional association approaches; however, GaMMA is designed such that it can use phase arrival time, phase-type identification, and amplitude information for association while simultaneously estimating the underlying event source characteristics (i.e., location and magnitude). Moreover, GaMMA does not require extra association steps of grid-search or supervised training. These attributes make GaMMA an appealing approach to address the challenges arising in processing large numbers of automatic picks in earthquake monitoring workflows.

2. Method

The objective of earthquake phase association is to post-process a large collection of phases picked on individual seismic stations and cluster them into groups of seismic phases originating from a common earthquake event, so that subsequent earthquake characterization tasks can be performed on individual events. Arrival times of P and S phases from the same earthquake follow a hyperbolic moveout that is determined by the hypocentral distance and the Earth model (i.e., seismic wave speed). This moveout allows association algorithms to distinguish between phases from different earthquakes. In this work, we extend the association problem by using both phase arrival time and phase amplitude information. On average, the phase amplitude scales with earthquake magnitude and

21699356, 2022, 5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021JB023249 by Ogs Trieste Istituto Nazionale, Wiley Online Library on [05/10/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

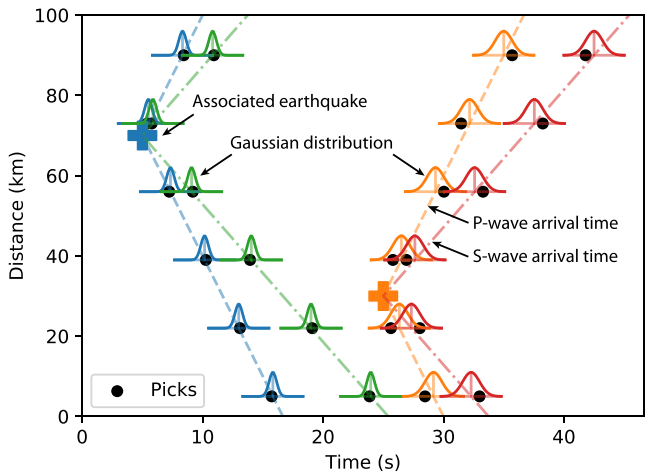

Figure 1. Gaussian mixture model for association. We model Gaussian distributions based on the theoretical phase arrival times and amplitudes as shown by the dashed lines from the two earthquakes (two crosses) in the figure, so that the probability of each pick is determined by the difference from the theoretical arrival time and amplitude. The phase association task, which is to estimate maximum likelihood of these picks, can be solved by the Expectation-Maximization (EM) algorithm. The EM iteration converges to the correct phase clustering for each earthquake and estimates its approximate source parameters. Note that we use both phase arrival time and phase amplitude information for phase association. For simplicity, only the time axis and a generalized distance from the edge are plotted.

decays with hypocentral distance. Thus, amplitude provides additional information to improve phase association. We formulate the association problem as follows: Given N seismic phases $(x_i, y_i, z_i, t_i, a_i)$, that is, arrival time t_i and amplitude a_i recored at the ith seismic station located at (x_i, y_i, z_i), we seek to group these phases into K earthquakes and estimate the underlying source parameters $(x_k, y_k, z_k, t_k, m_k)$, that is, location (x_k, y_k, z_k), origin time t_k, and magnitude m_k of the kth earthquake. We solve this association problem using the GMM (Permuter et al., 2006), which is an unsupervised clustering method that groups N data points into K clusters by maximizing the probability that these N data can be explained by a mixture of K Gaussian distributions. We incorporate the physical constraints on phase arrival time and amplitude into the GMM to make it suitable for our association problem. Figure 1 illustrates how we model the Gaussian distributions to calculate the probability of the sequence of phases generated by two causative earthquakes. The mathematical details are explained in the following sections.

2.1. Bayesian Gaussian Mixture Model

We cast the association problem in a probabilistic framework where we use a mixture of Gaussian distributions to model the probability of each phase pick generated by a mixture of earthquake sources:

$$p(\mathbf{x}_i) = w_i \sum_{k=1}^{K} \phi_k \mathcal{N}\left(\mathbf{x}_i | \mu_k, \mathbf{\Lambda}_k^{-1}\right) \tag{1}$$

$$\mathcal{N}\left(\mathbf{x}_i | \mu_k, \mathbf{\Lambda}_k^{-1}\right) = \frac{1}{(2\pi)^{n/2}} |\mathbf{\Lambda}_k|^{1/2} \exp\left(-\frac{1}{2}(\mathbf{x_i} - \mu_k)^T \mathbf{\Lambda}_k (\mathbf{x_i} - \mu_k)\right) \tag{2}$$

$$\sum_{k=1}^{K} \phi_k = 1 \tag{3}$$

where $\mathbf{x}_i$ represents a phase pick including arrival time and amplitude (t_i, a_i) values at the ith station. ϕ_k is the mixture component coefficient of the kth earthquake. $\mathcal{N}$ represents a Gaussian distribution, with $\boldsymbol{\mu}_k$ as its mean. $\boldsymbol{\mu}_k$ represents the theoretical phase arrival time and amplitude $\left(\hat{t}_{ik}, \hat{a}_{ik}\right)$ for each ith station determined by the kth earthquake. $\mathbf{\Lambda}_k$ is the precision (inverse covariance) matrix of the Gaussian distribution. w_i is the phase picking quality score between [0, 1]. n is the number of feature dimensions, which is one if only time information is used or two if both time and amplitude information are used. Based on the Gaussian mixture distribution, we can calculate the probability of a set of recorded phases $(x_1, x_2, \ldots, x_N)$ being generated by a mixture of K events. If we assume these observations are independent and identically distributed (i.i.d.), then the log likelihood function is given by:

$$\log\left(p(\mathbf{X}|\phi, \mu, \mathbf{\Lambda})\right) = \sum_{i=1}^{N} \log\left(w_i \sum_{k=1}^{K} \phi_k \mathcal{N}\left(\mathbf{x}_i | \mu_k, \mathbf{\Lambda}_k^{-1}\right)\right) \tag{4}$$

we find the assignment from N phase picks to K earthquakes, which is the goal of association, and the corresponding earthquake source parameters by maximizing the log likelihood of Equation 4.

A limitation of the GMM formulation is that we need to assume the number of underlying earthquakes K. To address this unknown, we implement the Bayesian GMM (Bishop, 2006), which uses variational inference to calculate approximate posterior distributions for the parameters of a Gaussian mixture distribution. Three conjugate priors are introduced. In particular, we use a Dirichlet prior for the mixture component coefficient, $p(\phi) = \mathcal{D}(\alpha_0)$ where α_0 is the weight concentration prior (Ferguson, 1973), which controls the concentration of mixture components; a Gaussian prior for the mean conditioned on the precision, $p(\mu_k|\Lambda_k) = \mathcal{N}\left(\mu_0, (\beta_0\Lambda_k)^{-1}\right)$ where μ_0 is the prior on the mean and β_0 is the precision prior on the mean; and a Wishart prior for the precision Λ_k, $p(\Lambda_k) = \mathbf{W}(\mathbf{W}_0, \nu_0)$ where $\mathbf{W}_0$ is the covariance prior and ν_0 is the degrees of freedom prior (Wishart, 1928),

21699356, 2022, 5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021JB023249 by Ogs Trieste Istituto Nazionale, Wiley Online Library on [05/10/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

which controls the estimation of covariance. The Bayesian model penalizes parameters that are away from the priors, which balances data fitting and model complexity. The mixture components (i.e., earthquakes) that do not contribute to explaining the data will have approximately zero mixture coefficients, so that we can choose a large number of components in the mixture model without over-fitting. In practice, we can initialize the space with many redundant earthquake hypocenters, and the Bayesian GMM suppresses unnecessary sources to infer an accurate number of earthquakes with associated picks.

2.2. Expectation-Maximization (EM) Algorithm

We use the the EM algorithm to solve for the maximum likelihood estimate of $p(\mathbf{x})$ (Equation 4). To consider physical constraints on phase arrival time and amplitude for association, we incorporate the estimate of earthquake location, origin time, and magnitude into the EM algorithm. We then iteratively update the assignments from picks to earthquakes in the E-step and optimize the earthquake parameters in the M-step:

2.2.1. E-Step

$$\gamma_{ik} = \frac{\phi_k \mathcal{N}\left(\mathbf{x}_i | \mu_k, \mathbf{\Sigma}_k\right)}{\sum_{k=1}^{K} \phi_k \mathcal{N}\left(\mathbf{x}_i | \mu_k, \mathbf{\Sigma}_k\right)} \tag{5}$$

where γ_{ik} is the probability that phase pick $\mathbf{x}_i$ (arrival time t_i and wave amplitude a_i) is generated by the kth earthquake.

2.2.2. M-Step

1. Effective number of picks assigned to the k-th earthquake

$$N_k = \sum_{i=1}^{N} \gamma_{ik} \tag{6}$$

$$\phi_k = \frac{N_k}{N} \tag{7}$$

2. Earthquake location, origin time, and magnitude of the k-th earthquake

$$\underset{(x_k, y_k, z_k, t_k)}{\text{minimize}} l\left(x_k, y_k, z_k, t_k\right) = \sum_{i=1}^{N} \gamma_{ik} \mathcal{L}\left(t_i, \hat{t}_{ik}\left(x_k, y_k, z_k, t_k\right)\right) \tag{8}$$

$$m_k = \frac{1}{N_k} \sum_{i=1}^{N} \gamma_{ik} \mathcal{F}_a'\left(a_i, d_{ik}\right) \tag{9}$$

3. Theoretical arrival time, amplitude, and statistics of residuals

$$\mu_k = \begin{bmatrix} \hat{t}_{ik} \\ \hat{a}_{ik} \end{bmatrix} = \begin{bmatrix} \mathcal{F}_t\left(x_k, y_k, z_k, t_k\right) \\ \mathcal{F}_a\left(m_k, d_{ik}\right) \end{bmatrix} \tag{10}$$

$$\mathbf{\Lambda}_k^{-1} = \frac{1}{N_k} \sum_{i=1}^{N} \gamma_{ik} \left(\mathbf{x}_i - \mu_k\right)\left(\mathbf{x}_i - \mu_k\right)^T \tag{11}$$

where $\mathcal{L}$ is a loss function of the absolute residuals between the picked phase arrival times t_i and the theoretical arrival times $\hat{t}_{ik}$ from the kth earthquake. Minimization of the loss function l gives an estimate of the earthquake location and origin time (x_k, y_k, z_k, t_k). m_k is the magnitude of the kth earthquake. $\mathcal{F}_t$ represents the function used to calculate theoretical phase arrival time $\hat{t}_{ik}$. $\mathcal{F}_a$ represents the function to calculate theoretical phase amplitude $\hat{a}_{ik}$ based on earthquake magnitude m_k, and $\mathcal{F}_a'$ represents the function to estimate earthquake magnitude using phase amplitude. d_{ik} is the distance from the kth earthquake to ith seismic station. Here we decouple the optimization of

21699356, 2022, 5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021JB023249 by Ogs Trieste Istituto Nazionale, Wiley Online Library on [05/10/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

earthquake magnitude (Equation 9) from the optimization of earthquake location and time (Equation 8). We use arrival times to constrain earthquake location and use phase amplitudes to constrain earthquake magnitude. Note that although we decouple the two optimizations, the precision matrix (Equation 11) considers the correlation between arrival time and amplitude residuals. In this way, both arrival time and amplitude information are used for the association process of clustering picks among earthquakes (Equation 5).

For the Bayesian GMM, we add another stage in the M-step to update the posterior parameters:

$$\alpha_k = \alpha_0 + N_k \tag{12}$$

$$\beta_k = \beta_0 + N_k \tag{13}$$

$$\mathbf{m}_k = \frac{1}{\beta_k}\left(\beta_0 \mathbf{m}_0 + N_k \mu_k\right) \tag{14}$$

$$\mathbf{W}_k^{-1} = \mathbf{W}_0^{-1} + N_k \mathbf{\Lambda}_k^{-1} + \frac{\beta_0 N_k}{\beta_0 + N_k}\left(\mu_k - \mathbf{m}_0\right)\left(\mu_k - \mathbf{m}_0\right)^{\mathrm{T}} \tag{15}$$

$$\nu_k = \nu_0 + N_k \tag{16}$$

The E-step is modified as:

$$\gamma_{ik} \propto \tilde{\pi}_k \tilde{\Lambda}_k^{1/2} \exp\left\{-\frac{D}{2\beta_k} - \frac{\nu_k}{2}\left(\mathbf{x}_n - \mathbf{m}_k\right)^{\mathrm{T}} \mathbf{W}_k \left(\mathbf{x}_n - \mathbf{m}_k\right)\right\}, \quad \sum_{k=1}^{K} \gamma_{ik} = 1 \tag{17}$$

$$\ln \tilde{\Lambda}_k = \sum_{i=1}^{D} \psi\left(\frac{\nu_k + 1 - i}{2}\right) + D \ln 2 + \ln \left|\mathbf{W}_k\right| \tag{18}$$

$$\ln \tilde{\pi}_k = \psi\left(\alpha_k\right) - \psi\left(\widehat{\alpha}\right) \tag{19}$$

where $\widehat{\alpha} = \sum_k \alpha_k$ and ψ is the digamma function (Abramowitz & Stegun, 1964). The explanation of these updating rules is detailed in Bishop (2006)'s textbook.

2.3. Earthquake Location and Magnitude Estimation

The iteration of the EM algorithm updates the clustering of picks based on earthquakes and optimizes the corresponding earthquake source parameters. We focus on efficient association rather than accuracy of earthquake source parameters, which can be realized once phases are properly associated, so we choose two basic approaches to estimate approximate earthquake locations and magnitudes. We optimize Equation 8 with a Huber loss function (Huber, 1992) as the target to reduce the effect of outliers:

$$\mathcal{L}_\delta\left(t - \hat{t}\right) = \begin{cases} \frac{1}{2}\left(t - \hat{t}\right)^2 & \text{for } \left|t - \hat{t}\right| \le \delta \\ \delta\left(\left|t - \hat{t}\right| - \frac{1}{2}\delta\right), & \text{otherwise.} \end{cases} \tag{20}$$

where the hyper-parameter δ is set to 1s in this test. For this proof-of-concept study, we use a uniform velocity model to calculate the theoretical phase arrival time:

$$\widehat{t}_{ik}\left(x_k, y_k, z_k, t_k\right) = \mathcal{F}_t\left(x_k, y_k, z_k, t_k\right) = \frac{d_{ik}}{v} + t_k \tag{21}$$

we then solve the minimization of Equation 8 using the BFGS algorithm (Fletcher, 2013). Advanced earthquake location algorithms and complex velocity models can also be applied to solving Equation 8 but at a higher computational cost.

To estimate earthquake magnitude in Equation 9, we use a linear relationship between the logarithm of the phase amplitude and earthquake magnitude:

$$\widehat{m}_{ik} = \mathcal{F}_a'\left(a_i, d_{ik}\right) = c_0 + c_1 \log a_i + c_2 \log d_{ik} \tag{22}$$

21699356, 2022, 5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021JB023249 by Ogs Trieste Istituto Nazionale, Wiley Online Library on [05/10/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Station correction terms can be added to consider site effects, such as site amplification factors (Münchmeyer et al., 2020). Based on the measured phase amplitude type, for example, displacement, peak ground velocity (PGV), or peak ground acceleration, we can choose from among the Richter empirical magnitude relationship (Richter, 1935), the Richter simulation-based prediction (Al-Ismail et al., 2020), or ground motion prediction equations (GMPE) (Picozzi et al., 2018) for Equation 22.

3. Results

We demonstrate the performance of GaMMA first on two synthetic examples and then on six days of data from the 2019 Ridgecrest, California earthquake sequence.

3.1. Synthetic Test

We created two synthetic experiments to demonstrate the association results of GaMMA. In the first 1D synthetic experiment, we generated a sequence of both P- and S-phases from six earthquake events. To model the errors that exist in real data, we added a 0.5s random error in the phase arrival times and scaled the phase amplitude (PGV) by a random factor between 0.3 and 3. We further added 30% false positive picks at random times. In total, 178 P- and S-phase picks from 40 stations were used for association (left panels of Figure 2). We used a simplified ground motion prediction equation of PGV from Picozzi et al. (2018)'s work: $\log PGV = -2.175–1.68 \log R + 0.93\, M$, where R is hypocentral distance and M is earthquake magnitude. The ground truth result is shown in the middle panels of Figure 2. The symbol size represents the relative size of the phase amplitude and the earthquake magnitude. We conducted two association tests, using in the first test only the arrival time information (Figure 2a), and in the second test both the arrival time and amplitude information (Figure 2b). The same parameters and initialization were used for both cases. We initialized the earthquake locations at the center of the research area and uniformly distributed the earthquake origin times. The association results are shown in the right panels of Figure 2. At least five of the six true earthquakes are successfully associated in both tests. However, the sixth event in the lower right corner of Figures 2a and 2b [ii] (marked in brown) is successfully associated only when amplitudes are used in conjunction with arrival times (Figure 2b [iii]). Without amplitudes (Figure 2a [iii]) we find multiple incorrect event associations marked by colors pink and purple. This occurs because the P-phases of this event have arrival times that overlap with another distant event's moveout (the event marked in red) and are mistakenly associated with this distant event. The extra information provided by amplitude adds the necessary extra constraint on distance, in addition to time, that allows the sixth event to be associated correctly. The test with amplitude information also correctly estimates the earthquake magnitudes. This synthetic experiment demonstrates that GaMMA benefits from using both the time and amplitude information in the association process.

In the second synthetic experiment, we used the station locations of the 2019 Ridgecrest earthquake sequence and randomly sampled event locations to quantify the association performance of GaMMA. The station locations and the 50 event locations are shown in Figure S1 in Supporting Information S1. We also calculated the performance of another association method REAL (Zhang et al., 2019) for comparison. We first analyzed the association speed of GaMMA (Figure 3). Because GaMMA is based on the GMM and does not perform a grid-search, its computational time scales only with the number of picks times the number of events, while the computational time of REAL scales with the number of picks times the number of grid points used. In this test, REAL uses a total of 18,491 ($41 \times 41 \times 11$) grid points to achieve a spatial resolution of 3 km and a depth resolution of 2 km. As a result, GaMMA has a faster association speed than REAL. We then analyzed the association accuracy of GaMMA. We generated synthetic datasets by adding random Gaussian errors in phase time and amplitude measurements, false positive picks, and biased velocity models of P and S waves. The default standard deviations of Gaussian random errors in this test are 0.4s for phase time and 0.4 ($\log_{10}$ m/s) for phase amplitude in a log scale; the default ratio of false positive picks is 37.5%; and the biased velocity model is 90% of the true velocity model. The average origin time intervals of the 50 synthetic events are around 10s or 5s. The association performance of REAL, GaMMA, and GaMMA using only time information are summarized in Table 1. Here, we define the associated event that is within 3s and 15 km from the true event as true positive association to calculate precision, recall, and F1-score (Powers, 2020). Under this idealized synthetic setting, both REAL and GaMMA can associate these events successfully. GaMMA using only time information

21699356, 2022, 5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021JB023249 by Ogs Trieste Istituto Nazionale, Wiley Online Library on [05/10/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

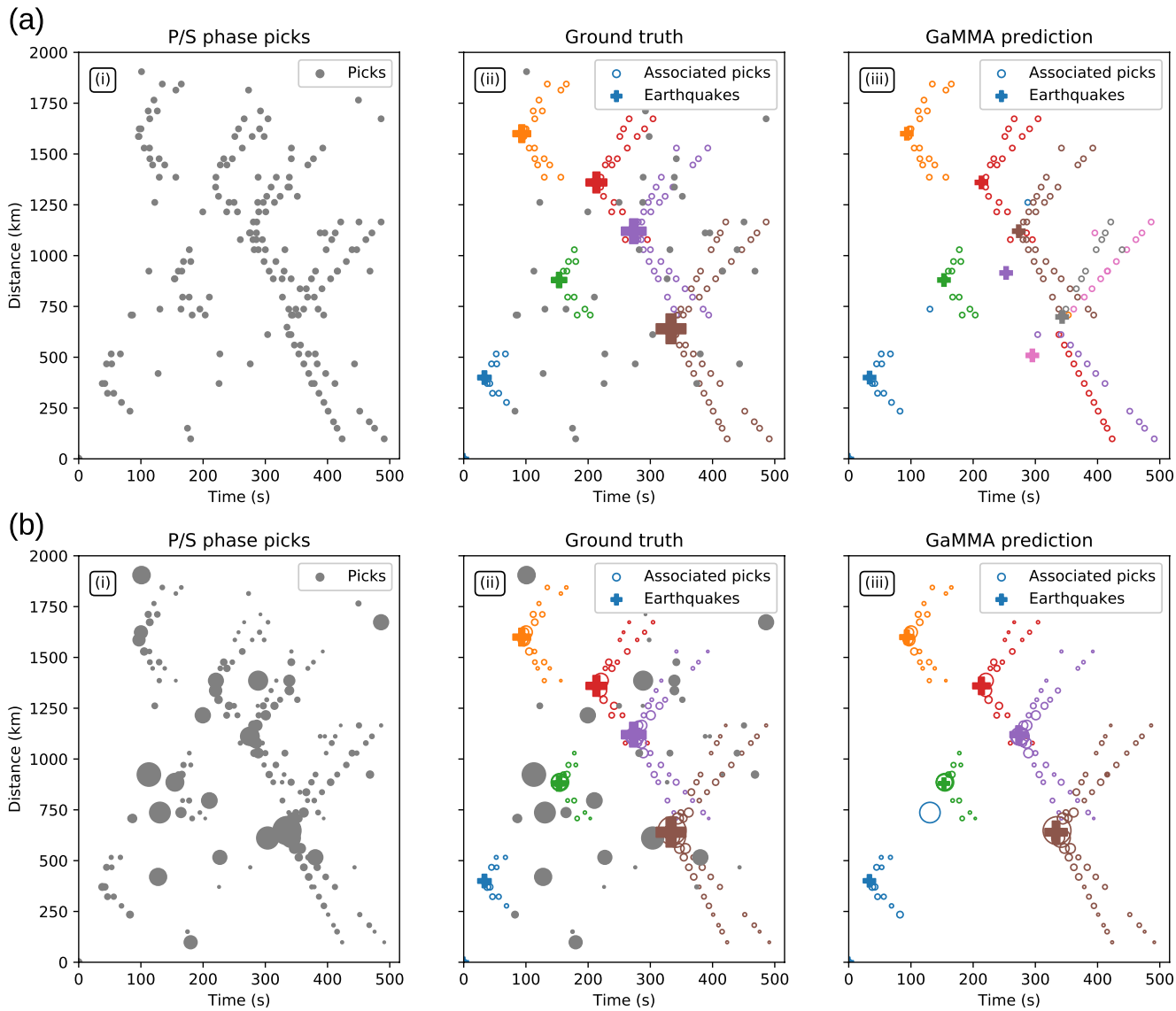

Figure 2. Synthetic example: (a) association using only time; (b) association using both time and amplitude. The left panels plot the P- and S-phase picks that scatter along the time axis and the distance from the edge of the region. The middle panels are plots of the ground truth association result. Unassociated false positives are plotted in gray. The circle size represents phase amplitude and the cross size represents earthquake magnitude. The right panels show the association results of the GaMMA method. *Note* that some phases in the lower right corner of panel (a) [iii] are mis-associated with another distant earthquake marked in red, because these phases can fit the moveout of both events. Amplitude information provides an extra constraint in distance that resolves this ambiguity as shown in panel (b) [iii].

has similar F1 scores as REAL, while GaMMA using both time and amplitude information achieves a better association result. In addition to these default values, we further conducted a sensitivity analysis of these three types of errors to analyze how GaMMA's association performance deteriorates with increasing errors (Figure S2–S4 in Supporting Information S1). For Gaussian random errors in phase measurements, GaMMA automatically adjusts the estimated covariance matrix (Equation 11), so that the estimated σ_{11} of phase time and σ_{22} of phase amplitude fall around the errors in the data (Figure S2 in Supporting Information S1). The association performance can be significantly impacted by false positive picks, which can lead to more false associations and low precision (Figure S3 in Supporting Information S1), and biased velocity models, which can cause systematic biases in theoretical phase times (Figure S4 in Supporting Information S1). Note that the errors in real data are much more complicated than our synthetic tests and may not follow a Gaussian distribution. This synthetic test is designed to compare the performance of GaMMA with REAL under the same setting and to analyze sensitivity under controlled error levels.

21699356, 2022, 5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021JB023249 by Ogs Trieste Istituto Nazionale, Wiley Online Library on [05/10/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

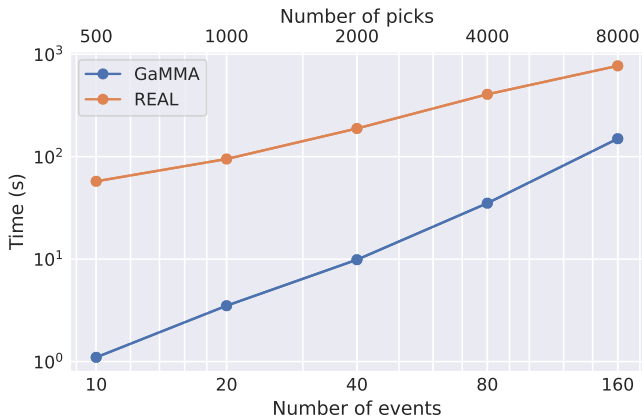

Figure 3. Comparison of association speed between REAL and GaMMA.

3.2. Test on the 2019 Ridgecrest Earthquake Sequence

We next applied GaMMA to part of the 2019 Ridgecrest earthquake sequence to evaluate its performance on real data. We focused on the initial 6 days of the sequence when a large number of earthquakes occurred and migrated from the southwest-striking fault to the northwest-striking fault. We applied the PhaseNet model (W. Zhu & Beroza, 2018) to extract picks from waveforms of "HH," "BH," "EH," and "HN" channels of 23 stations of the "CI" network within 1 degree of the location (−117.504 *W*, 35.705 *N*). We measured the PGV value over a 8s window after the phase arrival time. We then associated the detected 651,994 P-picks and 686,291 S-picks using GaMMA. We used a uniform velocity model with $v_p = 6$ km/s and $v_s = v_p/1.73$ for earthquake location estimation and a simple ground motion prediction equation as above for earthquake magnitude estimation. Because we used such a simple moveout behavior, we do not expect the earthquake locations to be accurate, but they are close enough for successful association given the relatively short source-receiver distances involved.

Figure 4 shows the statistics of the 24,327 associated earthquakes from 496,934 P-picks and 508,454 S-picks, leaving 155,060 P-picks and 177,837 S-picks unassociated. We also plot the 10,151 earthquakes in the SCSN catalog (SCEDC, 2013) for comparison. Both the associated earthquake locations and magnitudes agree with the SCSN catalog (Figures 4b and 4c). Figure 5 shows an association example with a dense sequence of picks occurring during a 6-min period. GaMMA associates 24 events during this period, while there are only 2 events in the SCSN catalog and 20 events in Ross, Idini, et al. (2019)'s template matching catalog. Figure 8 shows seismic waveforms of six newly detected events in Figure 5. We can see clear earthquake signals in these examples. These signals are relatively weak and can only be detected at a few stations. Figure 6 shows the residual distributions of the associated earthquake location and magnitude compared with the SCSN catalog. The statistics of mean, standard deviation (STD), and median absolute error (MAE) can be found in Table 2. The covariance matrix in Figure 6d shows that most of the associated earthquakes have small residuals of phase arrival time and amplitude, indicating that the phase picks match well with the theoretical values determined by the causative earthquakes found by association. These location and magnitude estimates can be further improved through the application of established earthquake location and magnitude algorithms once the picks have been associated.

Table 1
Comparison of Association Results Between REAL and GaMMA

	Synthetic data	Event interval	Precision	Recall	F1	σ_{11} (s)	σ_{22} ($\log_{10}$ m/s)
REAL	No errors	10s	0.94	0.90	0.92		
	+ phase measurement error + false positive picks	∼10s	0.71	0.70	0.71		
	+ velocity model error	∼5s	0.93	0.56	0.70		
GaMMA (time + amplitude)	No errors	∼10s	1.00	1.00	1.00	0.00	0.00
	+ phase measurement error + false positive picks	∼10s	0.92	0.94	0.93	0.52	0.39
	+ velocity model error	∼5s	0.94	0.94	0.94	0.54	0.41
GaMMA (time only)	No errors	∼10s	1.00	1.00	1.00	0.01	
	+ phase measurement error + false positive picks	∼10s	0.75	0.96	0.84	0.56	
	+ velocity model error	∼5s	0.53	0.72	0.61	0.57	

21699356, 2022, 5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021JB023249 by Ogs Trieste Istituto Nazionale, Wiley Online Library on [05/10/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

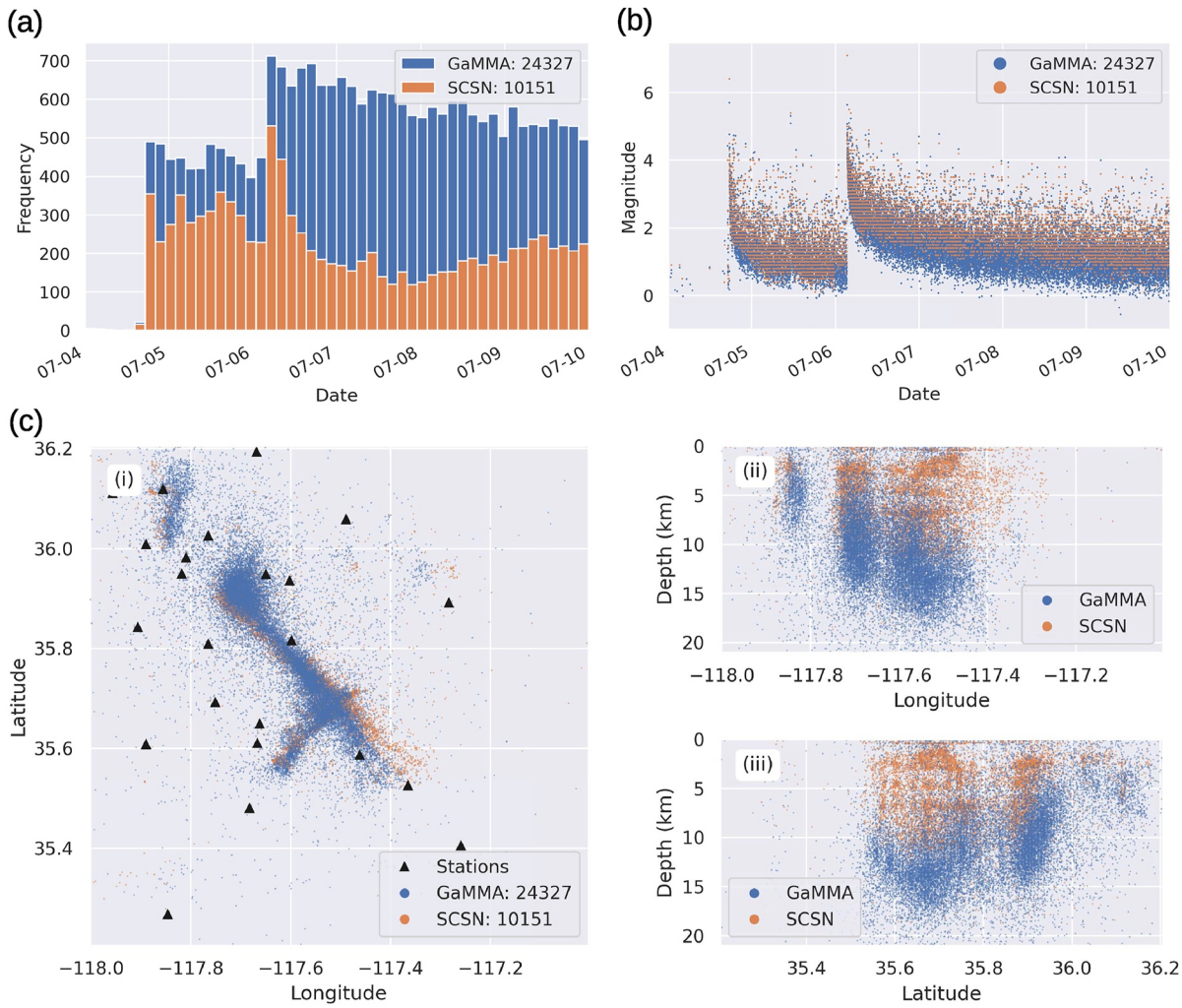

Figure 4. Association results of the Ridgecrest dataset: (a) associated earthquake frequency, (b) associated earthquake magnitude, (c) associated earthquake location. *Note* that because we use a uniform velocity model during association, we do not expect the earthquake locations to be accurate, but they are close enough for effective association.

We compared the catalog generated by GaMMA with three state-of-art catalogs (Liu et al., 2020; Ross, Idini, et al., 2019; Shelly, 2020). Table 3 shows the earthquake numbers in these catalogs during the same period. In each of these cases we assumed these catalogs as ground truth and analyzed whether the earthquakes they contain are also detected in GaMMA's catalog within a 5s window. Based on the recall rate, more than 90% of earthquakes in the catalogs of SCSN, Liu et al. (2020), and Shelly (2020) are successfully associated by GaMMA. The low precision and F1-score are due to the large number of new earthquakes associated by GaMMA. To verify whether these new earthquakes are reasonable, we compared the magnitude distributions of the four catalogs (Figure 7). Most of these new earthquakes associated by GaMMA have a small magnitude and follow the Gutenberg–Richter magnitude-frequency relationship (Gutenberg, 1956), which suggests that they may be legitimate detections of real earthquakes. Since we used a simple GMPE (Picozzi et al., 2018) to estimate earthquake magnitude based on PGV, we also see discrepancies at the high end of magnitude. The magnitude errors in Figure 6d indicates an underestimation of events above M5. Note that the associated earthquake numbers of these catalogs are dependent on the hyper-parameters used. For example, we applied two filtering criteria to remove low-quality associations: a minimum of 15 picks per event, $\sigma_{11} < 2.0$ s and $\sigma_{22} < 1.0$ ($\log_{10}$ m/s). Because of the

Table 2
Statistics of Residual Distributions in Figure 6

Error	Δx (km)	Δy (km)	Δz (km)	Δt (s)	Δm
Mean	−0.66	0.53	5.81	−0.37	−0.06
STD	2.55	2.21	4.15	0.91	0.23
MAE	2.03	1.64	6.35	0.68	0.15

21699356, 2022, 5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021JB023249 by Ogs Trieste Istituto Nazionale, Wiley Online Library on [05/10/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

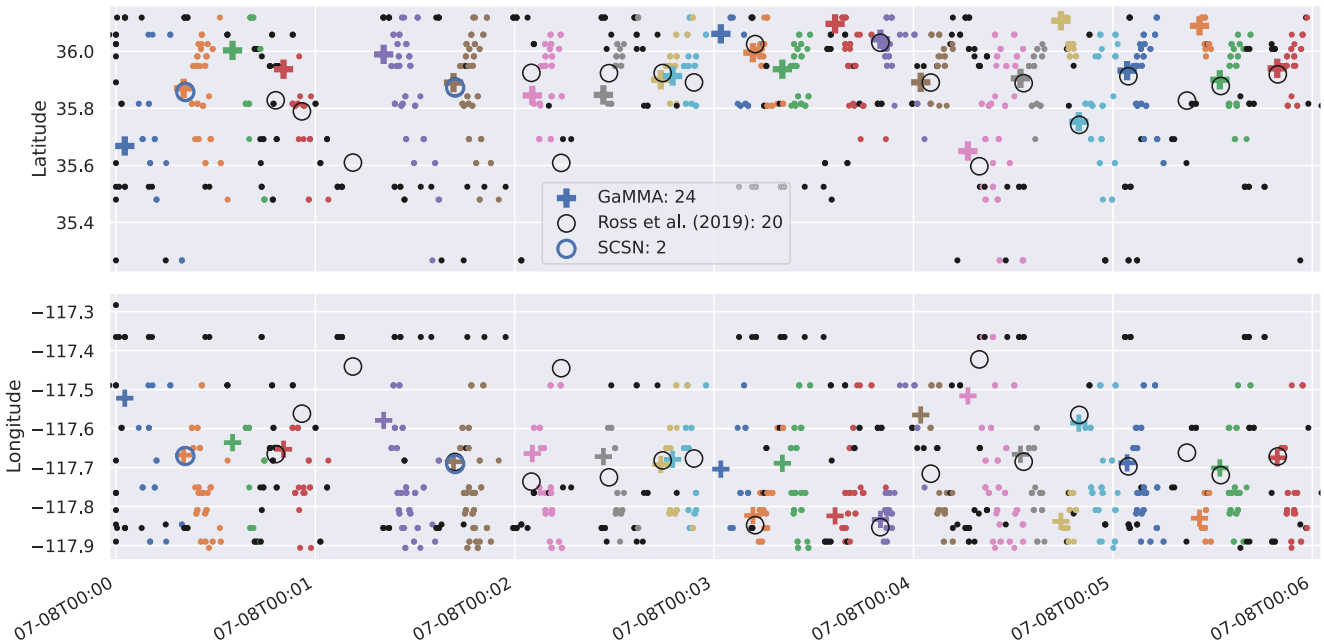

Figure 5. An example of association results from a dense sequence of phase picks starting at time 2019-07-08T00:00:00 (UTC). GaMMA associates 24 events during this period, while there are only 2 events in the SCSN catalog and 20 events in Ross, Idini, et al. (2019)'s template matching catalog.

unknown ground truth and the trade-off between the number of predictions and the false positives, it is challenging to evaluate the false positive associations in these catalogs.

4. Discussion

The GMM is an effective and widely used unsupervised learning method for clustering. We have combined the GMM with earthquake location and magnitude estimation to develop a novel GMM-based association method (GaMMA). We treat earthquake phase association as a clustering problem that aims to cluster phases based on the causative earthquakes. There are several advantageous features to approaching the phase association task as an unsupervised clustering problem. First, GaMMA optimizes the association result in a probabilistic framework, so that GaMMA can flexibly consider phase arrival time, phase amplitude, phase type, and pick quality in association and estimates the covariance of time and amplitude residuals. While it is difficult for conventional association methods to consider phase amplitude information, GaMMA can easily include phase amplitude information - both to improve association and to estimate earthquake magnitude. We can further add other useful information, such as back-azimuths, event duration, and S-P time lags, to the multi-variate Gaussian distribution (Equation 2) as extra constraints for association. Second, GaMMA does not require grid-search or training commonly used in other association methods. Supervised training methods could have poor generalization when applied to new regions with different station geometries and velocity models. Conventional grid-search methods require determining the grid size as an important hyper-parameter. A small grid size can scatter picks of a single earthquake into multiple grids; and the computational cost grows exponentially as grid size decreases; while a large grid size can associate multiple false-positive picks or picks from multiple earthquakes into the same grid leading to false associations. Last, in addition to processing archived seismic datasets, we can apply GaMMA in a real-time environment using a sliding window and associate picks inside the sliding window. The sliding window approach involves repeated associations of picks that remain in the sliding window, but because GaMMA is very efficient for short sequence data, the computational cost is not high. These strengths of GaMMA make it a promising approach for improved earthquake phase association, and hence improved earthquake monitoring in general.

We note that there are several limitations of GaMMA that need to be considered. First, the time complexity of the GMM scales with $O(K \cdot N)$, where K is the number of clusters (earthquakes) and N is the number of samples (phase picks), so the computational cost could become prohibitive for a long earthquake sequence. In

21699356, 2022, 5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021JB023249 by Ogs Trieste Istituto Nazionale, Wiley Online Library on [05/10/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

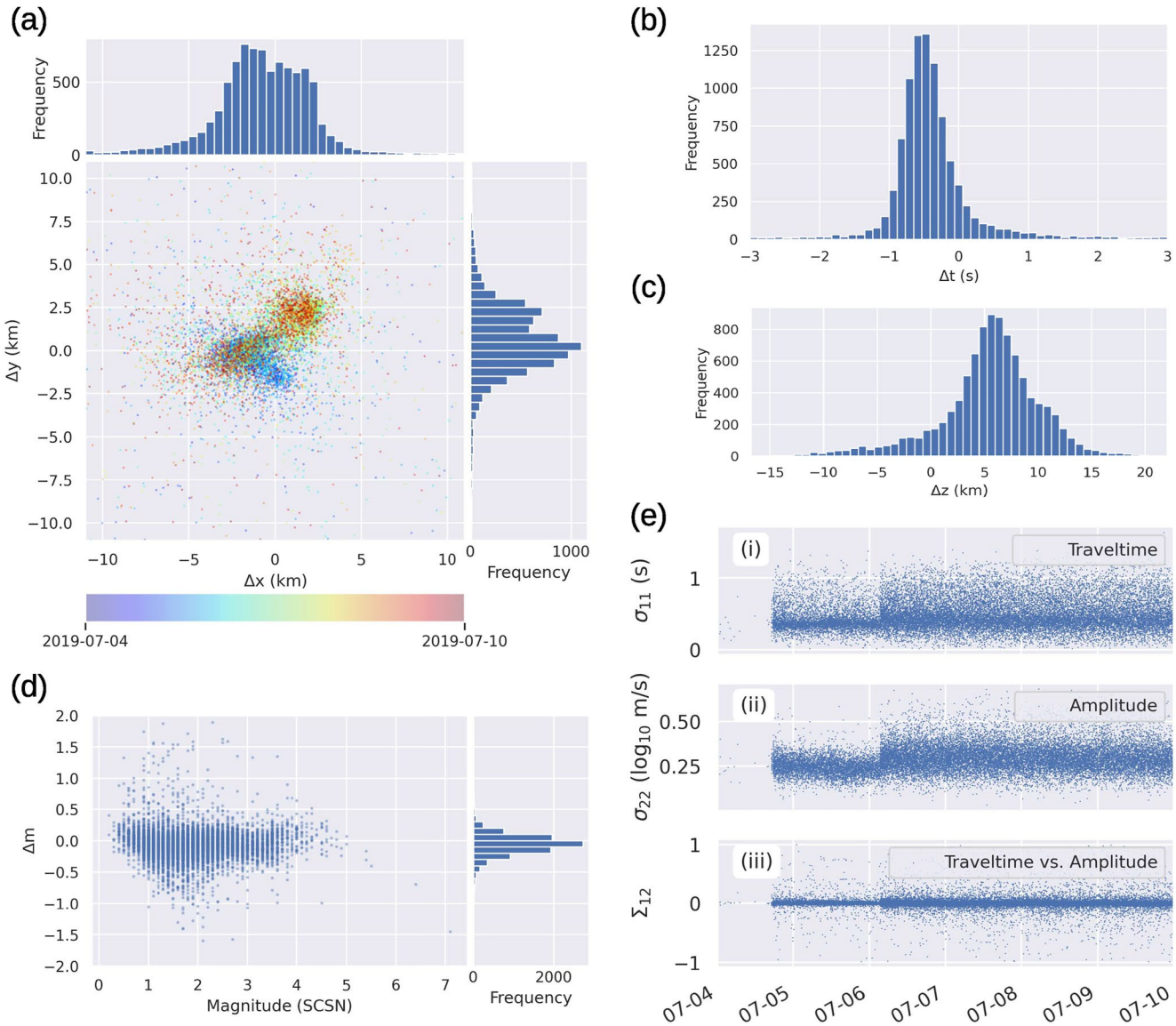

Figure 6. Residuals of (a) associated earthquake spatial location, (b) origin time, (c) depth, and (d) magnitude compared with the SCSN catalog. (e) The components of the covariance matrix of arrival time and amplitude residuals estimated by GaMMA.

practice, however, it is straightforward and effective to segment a long sequence into relatively shorter windows to improve the association speed by processing data in parallel, as phases that are separated by a certain time interval (depending on the source-receiver distance) cannot come from a single earthquake. In this work, we used the DBSCAN algorithm (Schubert et al., 2017) to divide picks into sub-windows for association. Second, the clustering results of the GMM are affected by the initialization state. In this work, we used a simple strategy to initialize the earthquake locations uniformly in time. The number of initialized earthquakes is proportional to an approximated number of earthquakes that is estimated by the number of phase picks divided by the number of stations. We define the ratio between initialized number and the approximated number of earthquakes as the oversampling ratio. The effect of this hyper-parameter is analyzed in Figure S5 in Supporting Information S1. Because we use a Bayesian GMM, we can use a larger number of initial clusters (i.e., earthquakes) for association. This simple strategy worked well in the experiments described above. Improving initialization strategies has the potential to improve the association performance further. Third, GaMMA ensures that one pick is only assigned to one earthquake, while conventional back-projection methods may attribute one pick to several earthquakes. But GaMMA does not consider station-based constraints, such as that

21699356, 2022, 5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021JB023249 by Ogs Trieste Istituto Nazionale, Wiley Online Library on [05/10/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

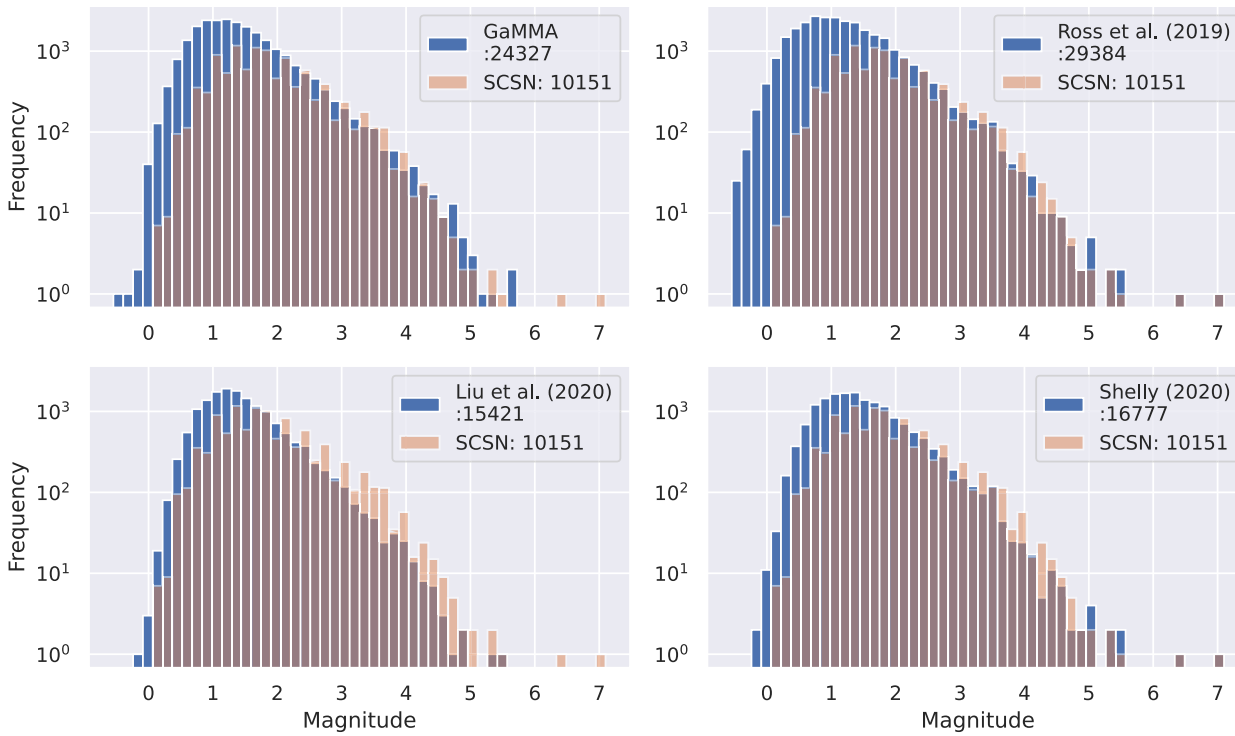

Figure 7. Comparison of earthquake magnitude distributions.

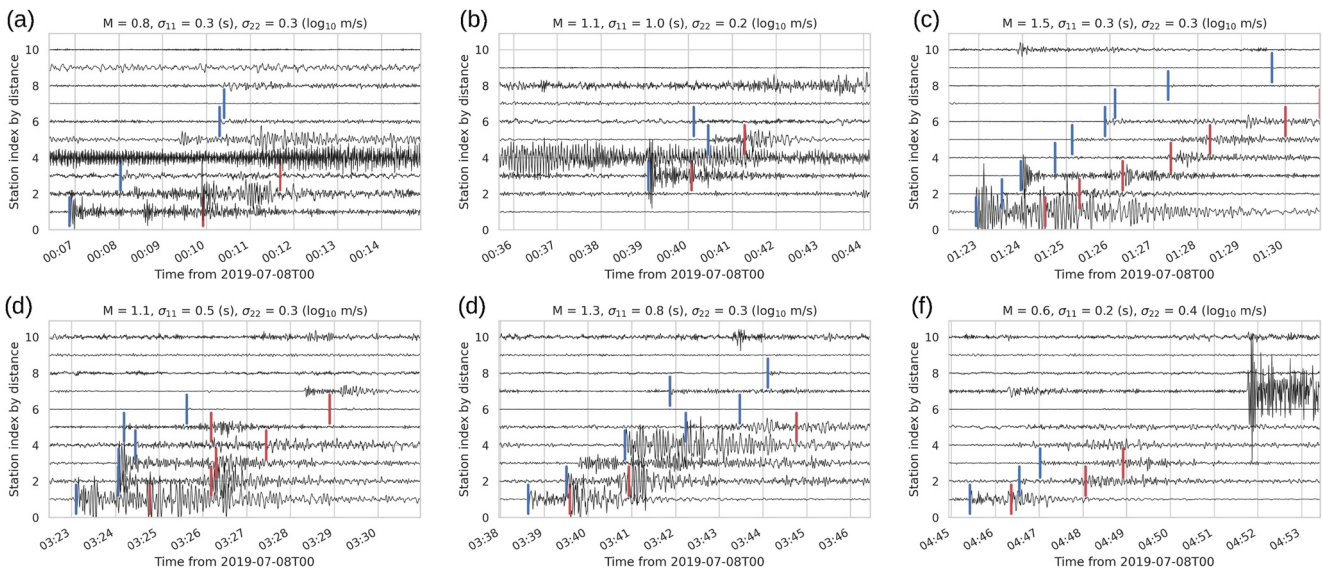

Figure 8. Waveforms of six newly detected events that are not Ross, Idini, et al. (2019)'s template matching catalog in Figure 5. M is the associated earthquake magnitude. σ_{11} and σ_{22} are the STD of associated phase time and phase amplitude respectively. These earthquakes have a small magnitude and can only be detected at a few stations.

only one P-pick or S-pick from each station is assigned to each earthquake. McBrearty, Gomberg, et al. (2019) accounts for this constraint using a constrained ILP (integer linear programming) solution in their association method, and there is a strong correspondence between that technique and our technique. One potential solution that would allow introduction of the station-based constraint is to add a normalizing scheme similar to Equation 5 over stations. We adopt a simple post-processing filter to remove multiple picks from the same

21699356, 2022, 5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021JB023249 by Ogs Trieste Istituto Nazionale, Wiley Online Library on [05/10/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 3
Comparison With Other Catalogs

Catalog		SCSN (10,151)	Ross et al. (2019) (29,384)	Liu et al. (2020) (15,421)	Shelly (2020) (16,778)
GaMMA (24,327)	Recall	0.94	0.61	0.95	0.89
	Precision	0.42	0.81	0.69	0.65
	F1-score	0.59	0.70	0.80	0.75

station and keep only the pick with the minimum arrival time residual for each associated earthquake. Finally, GaMMA assumes that the residuals of pick time or amplitude follow a Gaussian distribution. This assumption is not accurate for false positive picks. For example, false-positive picks are not randomly distributed and more likely to emerge from the coda of an event, other impulsive phases, or various noise sources. Adding a mixture component of background uniform distribution to account for false positive picks might be a helpful extension to be considered in future research (Melchior & Goulding, 2018). The amplitude measurement has the potential to be biased by events that happen closely in time, earthquake radiation patterns, site amplification factors, and other issues that affect amplitudes. Accurately measuring amplitude information for events occurring closely in time requires accurately detecting each event and then measuring the amplitude for the proper phases of each event. New algorithms for improving amplitude measurement could further improve the association performance of GaMMA.

5. Conclusions

We have developed a new phase association method, Bayesian Gaussian Mixture Model Association (GaMMA), which solves the phase association problem as an unsupervised clustering problem. To consider the physical constraints of phase arrival time and amplitude with earthquake location and magnitude, we incorporate optimization of earthquake location and magnitude into the EM algorithm that is commonly used for solving the GMM. GaMMA can use both arrival time and amplitude information to cluster picks from the same earthquake and simultaneously estimates both the earthquake location and magnitude from each cluster of picks. The experiment results for both synthetic tests and the 2019 Ridgecrest earthquake sequence demonstrate the effectiveness of GaMMA in associating a dense sequence of P- and S-phase picks under realistic conditions. The improved performance of GaMMA in association speed while using both time and amplitude information can associate more earthquakes from massive automatic phase pick datasets, thus enriching earthquake catalogs and improving earthquake monitoring.

Data Availability Statement

The phase picking and association data are available in Open Science Framework (https://doi.org/10.17605/OSF.IO/3GP72). The code is open source in GitHub (https://doi.org/10.5281/zenodo.6271310). Gaussian Mixture Model Association is developed based on the scikit-learn package (https://github.com/scikit-learn/scikit-learn). The data of 2019 Ridgecrest earthquake can be accessed from Southern California Earthquake Data Center.

Acknowledgments
This work is supported by the Department of Energy Basic Energy Sciences (DE-SC0020445).

References

Abramowitz, M., & Stegun, I. A. (1964). *Handbook of Mathematical Functions with Formulas, Graphs, and Mathematical Tables* (Vol. 55). US Government printing office.

Al-Ismail, F., Ellsworth, W. L., & Beroza, G. C. (2020). Empirical and synthetic approaches to the calibration of the local magnitude scale, ML, in southern Kansas. *Bulletin of the Seismological Society of America*, *110*(2), 689–697. https://doi.org/10.1785/0120190189

Allen, R. V. (1978). Automatic earthquake recognition and timing from single traces. *Bulletin of the Seismological Society of America*, *68*(5), 1521–1532. https://doi.org/10.1785/bssa0680051521

Arora, N. S., Russell, S., & Sudderth, E. (2013). NET-VISA: Network processing vertically integrated seismic analysis. *Bulletin of the Seismological Society of America*, *103*(2A), 709–729. https://doi.org/10.1785/0120120107

Barrett, S., & Beroza, G. (2014). An empirical approach to subspace detection. *Seismological Research Letters*, *85*, 594–600. https://doi.org/10.1785/0220130152

21699356, 2022, 5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021JB023249 by Ogs Trieste Istituto Nazionale, Wiley Online Library on [05/10/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Beroza, G. C., Segou, M., & Mostafa Mousavi, S. (2021). Machine learning and earthquake forecasting—Next steps. *Nature Communications*, *12*(1), 1–3. https://doi.org/10.1038/s41467-021-24952-6

Bishop, C. M. (2006). *Pattern recognition and machine learning*. Springer.

Dickey, J., Borghetti, B., Junek, W., & Martin, R. (2020). Beyond correlation: A path-invariant measure for seismogram similarity. *Seismological Research Letters*, *91*(1), 356–369. https://doi.org/10.1785/0220190090

Draelos, T. J., Ballard, S., Young, C. J., & Brogan, R. (2015). A new method for producing automated seismic bulletins: Probabilistic event detection, association, and location. *Bulletin of the Seismological Society of America*, *105*(5), 2453–2467. https://doi.org/10.1785/0120150099

Ferguson, T. S. (1973). A Bayesian analysis of some nonparametric problems. *Annals of Statistics*, 209–230. https://doi.org/10.1214/aos/1176342360

Fletcher, R. (2013). *Practical methods of optimization*. John Wiley & Sons.

Friberg, P., Lisowski, S., Dricker, I., & Hellman, S. (2010). Earthworm in the 21st century. In *EGU general assembly conference abstracts* (p. 12654).

Gibbons, S. J., Kværna, T., Harris, D. B., & Dodge, D. A. (2016). Iterative strategies for aftershock classification in automatic seismic processing pipelines. *Seismological Research Letters*, *87*(4), 919–929. https://doi.org/10.1785/0220160047

Gibbons, S. J., & Ringdal, F. (2006). The detection of low magnitude seismic events using array-based waveform correlation. *Geophysical Journal International*, *165*(1), 149–166. https://doi.org/10.1111/j.1365-246x.2006.02865.x

Gutenberg, B. (1956). The energy of earthquakes. *Quarterly Journal of the Geological Society*, *112*(1–4), 1–6. https://doi.org/10.1007/978-3-662-28668-5_1

Harris, D. B., & Dodge, D. A. (2011). An autonomous system for grouping events in a developing aftershock sequence. *Bulletin of the Seismological Society of America*, *101*(2), 763–774. https://doi.org/10.1785/0120100103

Hauksson, E., Yang, W., & Shearer, P. M. (2012). Waveform relocated earthquake catalog for southern California (1981 to June 2011). *Bulletin of the Seismological Society of America*, *102*(5), 2239–2244. https://doi.org/10.1785/0120120010

Huber, P. J. (1992). Robust estimation of a location parameter. In *Breakthroughs in statistics* (pp. 492–518). Springer. https://doi.org/10.1007/978-1-4612-4380-9_35

Klein, F. W. (2002). User's guide to HYPOINVERSE-2000, a fortran program to solve for earthquake locations and magnitudes. US Geological Survey. https://doi.org/10.3133/ofr02171

Liu, M., Zhang, M., Zhu, W., Ellsworth, W. L., & Li, H. (2020). Rapid characterization of the July 2019 Ridgecrest, California, earthquake sequence from raw seismic data using machine-learning phase picker. *Geophysical Research Letters*, *47*(4), e2019GL086189. https://doi.org/10.1029/2019gl086189

McBrearty, I. W., Delorey, A. A., & Johnson, P. A. (2019). Pairwise association of seismic arrivals with convolutional neural networks. *Seismological Research Letters*, *90*(2A), 503–509. https://doi.org/10.1785/0220180326

McBrearty, I. W., Gomberg, J., Delorey, A. A., & Johnson, P. A. (2019). Earthquake arrival association with backprojection and graph theory. *Bulletin of the Seismological Society of America*, *109*(6), 2510–2531. https://doi.org/10.1785/0120190081

Melchior, P., & Goulding, A. D. (2018). Filling the gaps: Gaussian mixture models from noisy, truncated or incomplete samples. *Astronomy and Computing*, *25*, 183–194. https://doi.org/10.1016/j.ascom.2018.09.013

Mousavi, S. M., Ellsworth, W. L., Zhu, W., Chuang, L. Y., & Beroza, G. C. (2020). Earthquake transformer—An attentive deep-learning model for simultaneous earthquake detection and phase picking. *Nature Communications*, *11*(1), 1–12. https://doi.org/10.1038/s41467-020-17591-w

Münchmeyer, J., Bindi, D., Sippl, C., Leser, U., & Tilmann, F. (2020). Low uncertainty multifeature magnitude estimation with 3-D corrections and boosting tree regression: Application to North Chile. *Geophysical Journal International*, *220*(1), 142–159. https://doi.org/10.1093/gji/ggz416

Park, Y., Mousavi, S. M., Zhu, W., Ellsworth, W. L., & Beroza, G. C. (2020). Machine-learning-based analysis of the Guy-Greenbrier, Arkansas earthquakes: A tale of two sequences. *Geophysical Research Letters*, *47*(6), e2020GL087032. https://doi.org/10.1029/2020gl087032

Patton, J. M., Guy, M. R., Benz, H. M., Buland, R. P., Erickson, B. K., & Kragness, D. S. (2016). *Hydra–the national earthquake information center's 24/7 seismic monitoring, analysis, catalog production, quality analysis, and special studies tool suite*. US Department of the Interior, US Geological Survey.

Peng, Z., & Zhao, P. (2009). Migration of early aftershocks following the 2004 Parkfield earthquake. *Nature Geoscience*, *2*(12), 877–881. https://doi.org/10.1038/ngeo697

Permuter, H., Francos, J., & Jermyn, I. (2006). A study of Gaussian mixture models of color and texture features for image classification and segmentation. *Pattern Recognition*, *39*(4), 695–706. https://doi.org/10.1016/j.patcog.2005.10.028

Picozzi, M., Bindi, D., Spallarossa, D., Di Giacomo, D., & Zollo, A. (2018). A rapid response magnitude scale for timely assessment of the high frequency seismic radiation. *Scientific Reports*, *8*(1), 8562. https://doi.org/10.1038/s41598-018-26938-9

Powers, D. M. (2020). *Evaluation: From precision, recall and F-measure to ROC, informedness, markedness and correlation*. arXiv preprint arXiv:2010.16061.

Reynolds, D. A., & Rose, R. C. (1995). Robust text-independent speaker identification using Gaussian mixture speaker models. *IEEE Transactions on Speech and Audio Processing*, *3*(1), 72–83. https://doi.org/10.1109/89.365379

Richter, C. F. (1935). An instrumental earthquake magnitude scale. *Bulletin of the Seismological Society of America*, *25*(1), 1–32. https://doi.org/10.1785/bssa0250010001

Ross, Z. E., Idini, B., Jia, Z., Stephenson, O. L., Zhong, M., Wang, X., et al. (2019). Hierarchical interlocked orthogonal faulting in the 2019 Ridgecrest earthquake sequence. *Science*, *366*(6463), 346–351. https://doi.org/10.1126/science.aaz0109

Ross, Z. E., Meier, M.-A., Hauksson, E., & Heaton, T. H. (2018). Generalized seismic phase detection with deep learning. *Bulletin of the Seismological Society of America*, *108*(5A), 2894–2901. https://doi.org/10.1785/0120180080

Ross, Z. E., Trugman, D. T., Azizzadenesheli, K., & Anandkumar, A. (2020). Directivity modes of earthquake populations with unsupervised learning. *Journal of Geophysical Research: Solid Earth*, *125*(2), e2019JB018299. https://doi.org/10.1029/2019jb018299

Ross, Z. E., Trugman, D. T., Hauksson, E., & Shearer, P. M. (2019). Searching for hidden earthquakes in southern California. *Science*, *364*(6442), 767–771. https://doi.org/10.1126/science.aaw6888

Ross, Z. E., Yue, Y., Meier, M.-A., Hauksson, E., & Heaton, T. H. (2019). PhaseLink: A deep learning approach to seismic phase association. *Journal of Geophysical Research: Solid Earth*, *124*(1), 856–869. https://doi.org/10.1029/2018jb016674

SCEDC. (2013). *Southern California Earthquake Center*. Caltech. [Dataset].

Schubert, E., Sander, J., Ester, M., Kriegel, H. P., & Xu, X. (2017). DBSCAN revisited, revisited: Why and how you should (still) use DBSCAN. *ACM Transactions on Database Systems*, *42*(3), 1–21. https://doi.org/10.1145/3068335

Seydoux, L., Balestriero, R., Poli, P., De Hoop, M., Campillo, M., & Baraniuk, R. (2020). Clustering earthquake signals and background noises in continuous seismic data with unsupervised deep learning. *Nature Communications*, *11*(1), 1–12. https://doi.org/10.1038/s41467-020-17841-x

21699356, 2022, 5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021JB023249 by Ogs Trieste Istituto Nazionale, Wiley Online Library on [05/10/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Shelly, D. R. (2020). A high-resolution seismic catalog for the initial 2019 Ridgecrest earthquake sequence: Foreshocks, aftershocks, and faulting complexity. *Seismological Research Letters*, *91*(4), 1971–1978. https://doi.org/10.1785/0220190309

Shelly, D. R., Beroza, G. C., & Ide, S. (2007). Non-volcanic tremor and low-frequency earthquake swarms. *Nature*, *446*(7133), 305–307. https://doi.org/10.1038/nature05666

Tan, Y. J., Waldhauser, F., Ellsworth, W. L., Zhang, M., Zhu, W., Michele, M., et al. (2021). Machine-learning-based high-resolution earthquake catalog reveals how complex fault structures were activated during the 2016–2017 central Italy sequence. *The Seismic Record*, *1*(1), 11–19. https://doi.org/10.1785/0320210001

Waldhauser, F., & Schaff, D. P. (2008). Large-scale relocation of two decades of northern California seismicity using cross-correlation and double-difference methods. *Journal of Geophysical Research*, *113*(B8). https://doi.org/10.1029/2007jb005479

Weber, B., Becker, J., Hanka, W., Heinloo, A., Hoffmann, M., Kraft, T., et al. (2007). SeisComP3—Automatic and interactive real time data processing. In *Geophysical Research Abstracts* (Vol. 9).

Wishart, J. (1928). The generalised product moment distribution in samples from a normal multivariate population. *Biometrika*, 32–52. https://doi.org/10.1093/biomet/20a.1-2.32

Woollam, J., Rietbrock, A., Leitloff, J., & Hinz, S. (2020). HEX: Hyperbolic event eXtractor, a seismic phase Associator for highly active seismic regions. *Seismological Research Letters*, *91*(5), 2769–2778. https://doi.org/10.1785/0220200037

Yeck, W. L., Patton, J. M., Johnson, C. E., Kragness, D., Benz, H. M., Earle, P. S., et al. (2019). GLASS3: A standalone multiscale seismic detection associator. *Bulletin of the Seismological Society of America*, *109*(4), 1469–1478. https://doi.org/10.1785/0120180308

Yoon, C. E., O'Reilly, O., Bergen, K. J., & Beroza, G. C. (2015). Earthquake detection through computationally efficient similarity search. *Science Advances*, *1*(11), e1501057. https://doi.org/10.1126/sciadv.1501057

Zhang, M., Ellsworth, W. L., & Beroza, G. C. (2019). Rapid earthquake association and location. *Seismological Research Letters*, *90*(6), 2276–2284. https://doi.org/10.1785/0220190052

Zhu, L., Chuang, L., McClellan, J. H., Liu, E., & Peng, Z. (2021). A multi-channel approach for automatic microseismic event association using RANSAC-based arrival time event clustering (RATEC). *Earthquake Research Advances*, 100008. https://doi.org/10.1016/j.eqrea.2021.100008

Zhu, W., & Beroza, G. C. (2018). PhaseNet: A deep-neural-network-based seismic arrival-time picking method. *Geophysical Journal International*, *216*(1), 261–273. https://doi.org/10.1093/gji/ggy423

21699356, 2022, 5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021JB023249 by Ogs Trieste Istituto Nazionale, Wiley Online Library on [05/10/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License